

PROJECT TITLE

Cyber Crime Case Management System

Student/Team Information

Team Name: *Team #*	*Team Avengers* *DSCPP-III-2026-T093*
Team Member 1 (Team Lead) *Last Name, Name: Sapra, Kusum* *Student ID: 2510011573* *Email: 2510011573@geu.ac.in*	
Team Member 2 *Last Name, Name: Singhal, Ananya* *Student ID: 2510370189* *Email: 2510370189@geu.ac.in*	
Team Member 3 *Last Name, Name: Garg, Rudransh* *Student ID: 2510370238* *Email: 2510370238@geu.ac.in*	
Team Member 4 *Last Name, Name: Gusain, Aryan* *Student ID: 2510370533* *Email: 2510370533@geu.ac.in*	

PHASE-II PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract (2 pts)

The Cyber Crime Case Management System keeps all the cybercrime case information in one place. It stores case details, victims, officers, evidence, investigation records, and reports. We wrote the project in C and C++ used data structures to manage and search records, and used MySQL as the database. We also made a backend/API that connects the application to the database. We have finished the main parts of our project,and they are working together.We can run the system and show how it works using the terminal. For the final phase, we want to create a simple web frontend and connect it to backend.

Updated Project Approach and Architecture (2 pts)

We split the project into separate modules so every part was easy to creste and test without depending on the rest. The C module uses linked lists, queues, priority queues, Stacks, and hash tables to store case records, handle cases in prior , keeps the investigation history, and search by Case ID. The C++ module has classes for cases, users, officers and investigations and uses OOP concepts like encapsulation and inheritance, the Officer class inherit from the User . We use MySQL to store the project information, and the backend/API connect the database with main features of system.We add validation, authentication, and error handling .Keeping the modules separate make the system easy to manage and make changes when needed. Next we will create a web frontend and connect it to backend.

Tasks Completed (7 pts)

Task Completed	Team Member
Requirement analysis, project planning, system architecture.	Kusum Sapra (Team Lead)
Buit the C++ classes for cases and users, officer inheritance, and evidence, investigation, and report modules and implementation of Linked List, Queue, Priority Queue, Stack, and Hash Table.	Ananya Singhal
MySQL database design, database tables and relationships, database connectivity, case CRUD operations, and officer management.	Rudransh Garg
System testing, debugging, module integration, project documentation, and final validation.	Aryan Gusain
Integration testing and checking that the main modules work together.	All members

Challenges/Roadblocks (7 pts)

During the development of our project, we faced a few technical issues. The problem was connecting our C++ program to MySQL.We face few issues when setting the compiler and libraries, but after correcting the build settings, header files, and database connection,and after fix all the things we get system working . We faced some difficulties when connect the C, C++, and database parts because each work different. To make things easy, we keep the modules separate and connect to make sure everything worked together. We ran into some linker errors because few object files have same names, which take some efforts to fix. Compiling the files separately with unique names fixed it. On the backend, we had to double-check API routing, validation, authentication and error handling. Testing each module by itself before running the system helps us find and fix errors more quickly .

Tasks Pending (7 pts)

Task Pending	**Team Member (to complete the task)**
Design and develop the web frontend and dashboard	Rudransh Garg and Ananya Singhal
Create case, officer, evidence, and investigation interfaces	Rudransh Garg and Ananya Singhal
Connect the frontend with the backend APIs	Kusum Sapra and Aryan Gusain
Complete integration testing, final validation, documentation, and deployment	All members

Project Outcome/Deliverables (2 pts)

We have a working system for manage cybercrime related information. It includes data structures to manage and search records, C++ classes for cases and related entities, a MySQL database, and a backend/API to connect everything and make data available to the application. We have create the main operations such as case management, officer assignment, evidence handling, investigation records, and report management. Working on the project help us understand how programming and database can be used to solve a real problem. The main thing is to create a web frontend and connect it to the backend we already have.

Progress Overview (2 pts)

We have completed most of the work,including the data structures, C++ OOP classes, MySQL database, and backend/API. All the modules are now connected and we can run the system through the terminal and check if everything is working as it should. We have test the main features and fix the issues that come up when we connect different parts. The web frontend is still incomplete. Connect frontend with the backend, test the user interface, and make application ready for users are the main task left. In the final stage, we will create a simple and easy to use interface and connect to the backend.We will then test the complete application to check that its features work well.Once everything is running properly,we finish the final project report and complete all the documentation.

Codebase Information (2 pts)

GitHub Repository: `https://github.com/kusumsapra/CYBER_CRIMECASE_MANAGEMENT`

Branch: `master`

The GitHub repository contains all the main project files,including the C data structure modules, C++ classes, database files, and backend code. We have completed the data structures, developed the C++ classes, set up MySQL database, and connect the application to database.We have also worked on the case and officer management features and connected the backend/API. We will continue using this repository to update the code while creating the frontend and test the complete application.

Testing and Validation Status (2 pts)

Test Type	**Status (Pass/Fail)**	**Notes**
Data Structures	Pass	Linked List, Queue, Priority Queue, Stack, and Hash Table operations tested
C++ OOP modules	Pass	Classes, methods, and officer inheritance tested
MySQL connection and CRUD	Pass	Database connection and main record operations tested
Backend/API	Pass	Major API endpoints, validation, and database communication tested
Web frontend	Pending	Planned for the final phase

Deliverables Progress (2 pts)

The main project components have completed,including requirement analysis, system design,data structures, C++ OOP modules, case and officer management, evidence and investigation modules, report management, and MySQL database integration.The backend/API, authentication and data validation, are complete and system has tested.

The remaining work is to create a simple and user-friendly web interface, connect to backend, and test the whole system to make sure everything works well. Once everything is working well, we will complete the final project documentation and get the application ready to use.